ZapBox Compact – Operating Instructions

Language: English | **Version:** oi940284

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Overview

The **ZapBox Compact** is an electronic switch for Bitcoin Lightning payments. A payment via the Lightning Network can switch an output – ideal for vending machines, presentations, event control, and many other applications.

Basic Equipment

Component	Description
Microcontroller	T-Display-S3 with 1.9" LCD display
Front panel	Available with 35° or 90° display front (90° also available for panel mounting)
Input	USB-C (Power IN)
Output	USB-A socket
Control	Two on-board micro buttons
Switch 1	3-position slide switch (AUTO / OFF / ON)
Switch 2	2-position slide switch (Invert Output)
Option BTC Ticker	Activatable via the Web Installer
Option NFC Module	For Bolt Cards (NTAG424 DNA) and standard NTAG213/215/216 (with LNURL-withdraw)

Views

Image 1: Front view

Image 2: Rear view

Connections

Input – USB-C Socket for Power Supply (5V)

Power the device via the **Power IN** connector with a USB-C cable at **5 V DC (max. 5 A)**.

> **Note:** The Power IN connector is not "intelligent". Some chargers or power modules with a USB-C output do not recognize the ZapBox and will not supply power. In this case, use a **USB-A output** of the power supply or a different power source.

Input – USB-C Socket on the Microcontroller (Data Access, behind right side panel)

To read data from or transfer data to the device, connect the ZapBox to a computer or laptop:

1. On the **right side of the front panel**, there is a small, concealed flap.
2. Open the flap by sliding it to the right from below using a **narrow screwdriver**.
3. Connect a USB-C cable to the connector on the microcontroller underneath.

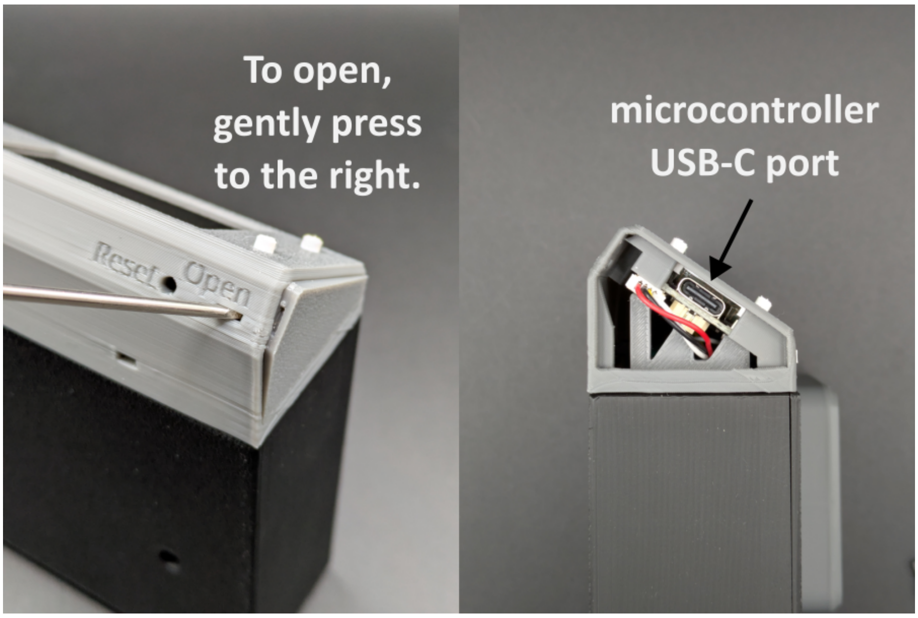

Image: Opening the panel and USB-C connector for data

> **Important Note:** The USB connector directly on the microcontroller is intended exclusively for flashing the firmware and transferring configuration parameters. During the flashing process, no load may be connected or switched at the output, as this can cause malfunctions or **damage to the microcontroller**.
>
> It is therefore recommended:
>
> - not to connect any load to the output during the USB connection, or
> - to additionally connect the regular **Power IN input** to the same power supply. This ensures that the current for the power relay does not flow through the microcontroller and overload it.

Output – USB-A Socket (Switched 5V Voltage)

The USB socket is switched via a relay switching contact. The **total load** of the sockets should **not exceed 3 A**.

Controls

The ZapBox has two small **on-board micro buttons** directly connected to the microcontroller. All functions are accessible via the micro buttons. Additionally, the ZapBox has a reset button on the underside of the front panel and two slide switches on the side.

Function Overview – Micro Buttons

Function	Micro Button
Show help page	Press HELP 1×
Next page / product change	Press NEXT 1×
Show REPORT page	Press HELP 2×
Enter config mode	Hold NEXT for at least 5 seconds

Function Overview – Slide Switches

The ZapBox Compact has two slide switches.

Switch 1 – Triple Slide Switch (AUTO / OFF / ON)

Position	Function
A (AUTO)	Automatic mode – Normal operation
0 (OFF)	Power supply interrupted – Output OFF
1 (ON)	Output permanently ON (with Switch 2 - Inverse = OFF)

Switch 2 – Dual Slide Switch (standard / inverted)

Position	Function
Std. (Standard)	The output is de-energized at rest (0 V). After switching, 5 V is present at the USB sockets.
Inv. (Inverse)	The output is at 5 V at rest. After switching, the output changes to 0 V (inverse switching).

> **Note:** If the dual switch is set to **Inv.** and the triple switch is in **position 1**, the output is – unlike in normal operation – **OFF** instead of ON.

Setup and Commissioning

The ZapBox is tested after manufacturing and delivered with the current firmware – however, it is not yet configured. The software is actively developed, so it is recommended to flash the ZapBox with the latest firmware right from the start and then perform a configuration. There is a convenient **Web Installer** for this purpose.

Here is a step-by-step guide for setup:

1. Open the right side panel of the front panel, as described above under "Input – USB-C Socket on the Microcontroller".
2. Connect the ZapBox to the USB-C port with a cable and connect it to a computer.
3. Open a Chromium browser, for example Google Chrome, Microsoft Edge, Brave, Vivaldi, Opera, or Helium.

4. Navigate to the Web Installer page **https://installer.zapbox.space/**.
5. Flash the latest "Latest" version, as described in step 1 of the Web Installer.
6. After flashing, close the small window and go to step 3 – Load config values. There, click the `Connect` button.
7. You should now see `Connected` and `Config mode` in the green field, provided the ZapBox is also in `SERIAL CONFIG MODE`. The display should show this. If not, check step 2 – Prepare connection.
8. The ZapBox requires three parameters: `WiFi SSID` / `WiFi password` / `Device settings string`. You get the device settings string from your LNbits wallet. Add the **Bitcoin Switch** or **ZapBox** extension for this. The ZapBox extension also supports the NFC module; otherwise they are identical.
9. After all three parameters have been entered, save them using the `Write Config` button and restart the ZapBox with the `Restart` button.

That should be all. After initialization, the ZapBox will display the QR code of the product and is ready for the first payment and subsequent action at the USB output.

In case of errors or malfunctions, please refer to the "Error Detection & Report" and "Troubleshoot" chapters further down on the Web Installer page.

Option: NFC Module

Depending on the configuration, an NFC module may be installed on the **top of the ZapBox**. Alternatively, the module is also available separately. The ZapBox currently supports the following card types:

- **Boltcards** (NTAG424)
- **LNURL-Withdraw** from NTAG21x (213 / 215 / 216)

The LNbits ZapBox Extension is required for this function. It must be supported by the LNbits server.

Technical Specifications

Property	Value
Supply voltage	5 V DC via USB-C
Maximum input current	5.0 A
Output power	max. 3.0 A (recommended)
Display	1.9" LCD (T-Display-S3)
Communication	Wi-Fi (ESP32-S3)
Payment protocol	Bitcoin Lightning Network

Safety Information

- Operate the device exclusively with the specified supply voltage.
- Do not exceed the maximum current loads of the outputs.
- Do not perform any work on the relay contacts under load.

- The device is not suitable for use in humid or wet environments.
- Keep out of reach of children.

Further Links

Resource	**Link**
Overview of all ZapBox models	https://zapbox.space/
Web Installer, quick overview & troubleshooting	https://installer.zapbox.space/
Detailed documentation (parameters & functions)	https://ereignishorizont.xyz/zapbox/
GitHub repository (software, PCB layouts, 3D print files, operating instructions)	https://github.com/AxelHamburch/ZapBox
ZapBox Extension	https://github.com/AxelHamburch/zapbox_extension
LNbits	https://lnbits.com/

Subject to changes and errors. As of: 2026